1. Clinical Question & Core Hypothesis

Clinical Question

Clinical trials combine pembrolizumab with chemotherapy (cisplatin-containing regimens) expecting TLS-rich tumors to benefit from enhanced T-cell immunity, yet KRAS-driven signaling and therapy-induced DNA damage could modulate this expected benefit. 2) Tier 1 paths used: I built on the KRAS → cell death / DNA-damage context and the cisplatin → mutation axis because they directly suggest mechanisms by which tumor-intrinsic KRAS activity could alter the antigenic consequences of cisplatin and thereby affect immune-mediated therapy. 3) Tier 2 evidence supporting the claim: the clinical-trial/pembrolizumab context and the notion that chemotherapy induces cell death and mutations support testing whether chemo-induced neoantigen generation or DNA-damage responses are altered by KRAS status. 4) Endpoint choice: I selected therapy_response because the path centers on pembrolizumab + chemotherapy combinations and the immediate translational question is who benefits from that regimen. 5) Exclusions: I deliberately excluded tissue-specific (colon) generalization and treating "Mutation" as the primary anchor to keep the hypothesis mechanistically sharp; those remain as secondary_modifiers for exploratory stratification?

Working Hypothesis

In TLS-rich tumors, activating KRAS mutations rewire the cisplatin-induced DNA-damage response and antigen processing pathways, leading to a qualitatively altered neoantigen landscape and reduced MHC-mediated presentation. This dampens TLS-driven T-cell activation and effector function, thereby reducing the efficacy of pembrolizumab plus cisplatin-containing chemotherapy compared with TLS-rich KRAS-wild-type tumors. The magnitude of this effect will correlate with measurable DDR/antigen-presentation perturbations (e.g., increased/f altered γH2AX signaling, reduced TAP1/MHC I expression) and with TLS-associated immune activity markers (lower IFN-γ signature, diminished TLS B/T cell activity) in KRAS-mutant versus KRAS-wild-type TLS-rich tumors.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	KRAS, cisplatin
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

KRAS: KRAS → cell death / DNA-damage context and the cisplatin → mutation axis because they directly suggest mechanisms by which tumor-intrinsic KRAS activity could alter the antigenic consequences of cisplatin and thereby affect immune-mediated therapy.

cisplatin: cisplatin → mutation axis because they directly suggest mechanisms by which tumor-intrinsic KRAS activity could alter the antigenic consequences of cisplatin and thereby affect immune-mediated therapy.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	pembrolizumab	Supporting modifier from knowledge graph
Secondary	Mutation	Supporting modifier from knowledge graph
Secondary	genetic mutations	Supporting modifier from knowledge graph
Secondary	colon	Supporting modifier from knowledge graph
Secondary	Tissue	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: in vitro KRAS-mutant vs WT tumor cell lines treated with cisplatin to assess DDR and antigen presentation;

Prediction 2: co-culture or functional assays with T cells to measure TLS-relevant cytotoxic responses;

Prediction 3: in vivo models engineered to form TLS and differing KRAS status with cisplatin ± PD-1 blockade to evaluate TLS architecture, T-cell activity, and tumor control;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- pembrolizumab: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- Mutation: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.